Cell dynamics on energy landscape: Comparing attractor detection in Boolean network and diffusion-based models under *in silico* gene perturbations

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Abstract

Steady-state attractor detection differs fundamentally between discrete Boolean network (BN) and continuous diffusion-based model because they operate on different state spaces and exhibit distinct dynamic behaviors. Here, we propose **CellLand** (Cell dynamics on energy Landscape) to systematically compare BN- and diffusion-based approaches for quantifying cellular dynamics on energy landscape, with a particular focus on attractor identification under *in silico* gene perturbations. Although the energy landscape provides a unifying conceptual framework, BN-based models face specific challenges, including state-space explosion and pronounced sensitivity to network architecture, when assessing attractor stability and transition accessibility. Nevertheless, BN-based analyses can remain reliable and interpretable when the underlying network topology is suitable, while being comparatively less sensitive to parameter-estimation uncertainty and technical noise. Importantly, we show that attractor identification depends critically on network structure, and that no general theory currently exists for optimizing BN topology for dynamic cell-state modeling. Our findings deepen the comparative understanding of BN- and diffusion-based frameworks and provide practical guidelines for selecting and designing models for robust simulation of complex gene-regulatory system under perturbations.

Keywords: Cell-state dynamics, Energy landscape, Gene regulatory network, Boolean network, Probabilistic diffusion equation, Gene perturbations

Introduction

Computational models of cell-state dynamics aim to represent, predict and ultimately steer cellular behavior *in silico* [1, 2]. By adopting a dynamical-system perspective, such models describe how cellular states emerge, stabilize, and transition over time, enabling systematic analysis of robustness and responses to genetic or environmental perturbations [2–4]. Within this framework, a steady state is a configuration in which all system variables remain constant over time in the absence of external perturbations [2]. Formally, it corresponds to a solution for which the time derivatives of all variables are zero, indicating that the system has reached dynamical equilibrium. Analyzing steady states is fundamental for understanding the long-term behavior and stability of dynamical systems. In biological contexts, steady states (attractors) often correspond to meaningful phenotypes, such as distinct cell fates or physiological equilibria [5, 6]. Identifying and characterizing these attractors, i.e., the steady states toward which the system naturally evolves, enables prediction of system responses to perturbations and elucidates possible transitions between different phenotypic and functional states [7].

Dynamical systems can be described using either discrete or continuous modeling frameworks. Boolean network (BN), first introduced by Kauffman in 1969 [8], is a commonly used discrete approach in which components assume binary values and are updated according to logical rules [9]. BN models have a lot of practical applications in different areas such as systems biology, computer science and engineering, see for instance Trairatphisan et al. [10], and are particularly effective for representing gene regulatory networks (GRNs) and cellular decision-making processes [11]. This success reflects that biological networks often depend on binary gene states rather than precise expression magnitudes [12, 13]. Consistent with this, binarized expression suffices for spatial clustering [14, 15] and pattern discovery [16, 17], validating Boolean abstractions of spatial transcriptomics. By contrast, continuous models employ real-valued variables and differential equations to describe the evolution of the system. Among them, the probabilistic diffusion equation (i.e., diffusion-based) framework is especially suitable for capturing stochastic dynamics and quantifying Waddington energy [5] or pseudo-energy landscapes in complex biological networks [18]. Quantifying the phenotypic landscape of perturbed networks and identifying driver genes (or master regulators) that restore the landscape toward that of a reference (nominal) network are crucial for recovering realistic biological behavior and guiding therapeutic interventions [19, 20].

Various modeling strategies have been used to study cellular phenotypic transitions by performing *in silico* gene perturbations on prescribed static GRNs [21], under the assumption that cellular identity is governed by complex regulation of gene expression [2]. BN-based models are extensively used in this context; for example, Font-Clos et al. applied a BN framework to investigate the landscape underlying epithelial-mesenchymal transition (EMT) plasticity, revealing multiple steady states associated with distinct phenotypes [22]. Their study illustrates key challenges of steady-state analysis in discrete systems, including the combinatorial explosion of possible network configurations and the sensitivity of attractor detection to network topology and update schemes. Similarly, continuous models, diffusion-based models governed by Fokker-Planck-type equations, provide powerful tools for characterizing energy landscapes and the dynamics of complex biological networks [23]. For instance, Kang and Li proposed a dimension-reduction strategy to study the energy landscape of a metabolism-EMT network, enabling identification of intermediate and steady states [24].

More recently, machine-learning-based methods have substantially expanded the computational toolbox for landscape inference and dynamical reconstruction. Deep-learning models such as EPR-Net have enabled the direct inference of non-equilibrium potential landscapes and steady-state entropy production from high-dimensional dynamics [25]. In parallel, Neural ODE and flow-matching models have been used to reconstruct continuous cell-state trajectories and latent dynamical laws from sparse single-cell data [26]. Related data-driven frameworks such as CellOracle infer gene-regulatory networks from single-cell multi-omics data and perform *in silico* transcription-factor perturbations to predict cell-state transitions [2]. Along a related geometric direction, GeoBridge models cellular dynamics on the transcriptional manifold and maps nonlinear trajectories to straight paths in a latent space, enabling biologically meaningful interpolation and reconstruction of unobserved intermediate states [27]. Together, these studies show that cellular state transitions can be investigated from discrete, diffusion-based, machine-learning, and geometry-informed perspectives, each offering complementary advantages for representing multistability, transition dynamics, and the organization of cellular-state landscapes.

From the perspective of statistical physics, the evolution of a system can be viewed as motion on an energy landscape along dynamical trajectories [28, 29]. When thermal fluctuations are strong enough to overcome the barriers separating local minima, the system can transition between distinct basins of attraction [30]. The smallest barrier between any two minima is associated with at least one intermediate, mixed configuration, which represents the most probable transition state visited as the system moves from one basin to the other [31]. Motivated by this viewpoint, we analyze three GRNs associated with hepatocellular carcinoma (HCC), melanoma, and a synthetic dataset generated by SERGIO [32]. For each network, we construct phenotypic/state landscapes by assigning pseudo-energy values to stable attractor states embedded in a principal component analysis (PCA)-reduced space. We then systematically compare the cell-state dynamics on energy landscapes obtained from a discrete BN description with those derived from a continuous diffusion-based formulation, thereby elucidating both the differences between these frameworks and the distinct challenges they face in identifying steady states, intermediate states, and transition pathways in complex gene regulatory system under *in silico* gene perturbation.

Results

Overview of identifying master genes driving cell-state transition dynamics In this study, we aim to identify master genes that drive cellular state transitions on a two-dimensional (2D) pseudo-energy landscape. Specifically, we seek mechanistic insight into the regulation of cell identity by evaluating how two dynamical systems models (BN and diffusion-based) capture phenotypic transitions induced by transcription factor (TF) perturbations. As an illustrative example, we first consider epithelial-mesenchymal transition (EMT) and its reverse process, mesenchymal-epithelial transition (MET), and then introduce our *in silico* perturbation strategy.

Fig. 1A schematically depicts EMT and MET. Epithelial (E) cells are typically non-motile, whereas mesenchymal (M) cells exhibit migratory and invasive properties. Because EMT is reversible, cells may also occupy intermediate hybrid E/M states that co-express epithelial and mesenchymal markers; on the pseudo-energy landscape, these correspond to mixed states located between the E and M basins. **Fig.** 1B outlines the *in silico* perturbation strategy used to test whether a given GRN can recapitulate TF perturbation-induced changes in cell phenotype or fate following TF knockdown (KD) or overexpression (OE). The GRN is modelled as a directed, weighted TF-target gene (TG) network. Perturbation of a TF is simulated by fixing its activity and propagating its influence to downstream TGs along outgoing edges, with the magnitude of the effect determined by the corresponding edge weights. Following Peng et al. [33], TF KD is implemented by setting the expression of the perturbed TF to 0 in the diffusion model and to -1 in BN model, thereby defining an initial perturbed state vector for each cell. This state is then iteratively updated according to Eq. (2) for the diffusion model or Eq. (4) for the BN model until convergence. The resulting stable perturbed state yields a cell-specific perturbation vector that encodes the predicted direction of fate change (**Fig.** 1C, bottom). To enable direct application of the BN model to RNA-seq data, we use a minimal binarization procedure (**Fig.** 1C, top right). Reference E and M cell populations are first selected to define gene-specific expression thresholds from their respective expression distributions. For each gene i, expression values above the threshold are assigned $+1$ (expressed), whereas values below the threshold are assigned -1 (not expressed). Using this unified framework, we analyse three GRNs derived from HCC [22], melanoma [23], and SERGIO [32] datasets, whose key structural properties are summarized in Table 1. In all three networks, each node assumes a binary state in the BN model ($+1$ for expressed and -1 for not expressed) and a continuous expression value in the diffusion model (nonzero for expressed and 0 for not expressed).

Table 1: **Summary of the interaction networks analysed in this study.** Directed edges denoted by $\rightarrow$ indicate positive (activating) regulation from gene i to gene j, whereas edges denoted by $\dashv$ indicate negative (inhibitory) regulation.

	EMT-MET interaction network in HCC	**Gene regulatory network in melanoma**	**Signed interaction network in SERGIO**
# nodes	72	17	100
# edges	139	64	258
Positive regulatory ($\rightarrow$)	107	43	179
Negative regulatory ($\dashv$)	35	21	79
# self-loops	0	9	0
Avg. number of neighbors	3.694	5.294	5.160
Network diameter	13	4	2
Network radius	7	1	1
Characteristic path length	5.257	1.854	1.743
Clustering coefficient	0.400	0.291	0.012
Network density	0.027	0.202	0.026
Connected components	1	1	1
Multi-edge node pairs	6	10	0

Landscape of the EMT-MET regulatory network in HCC EMT operates in normal development and wound repair and is co-opted during tumor dissemination. To determine how E-M plasticity is encoded by regulatory interactions, we applied the BN model in Eq. (4) to the EMT-MET circuit in HCC (**Fig.** 1D) to reconstruct the corresponding phenotypic landscape. A two-dimensional principal component (PC) projection of 10^6 simulations, each initiated from a random state and evolved to steady state, resolved the

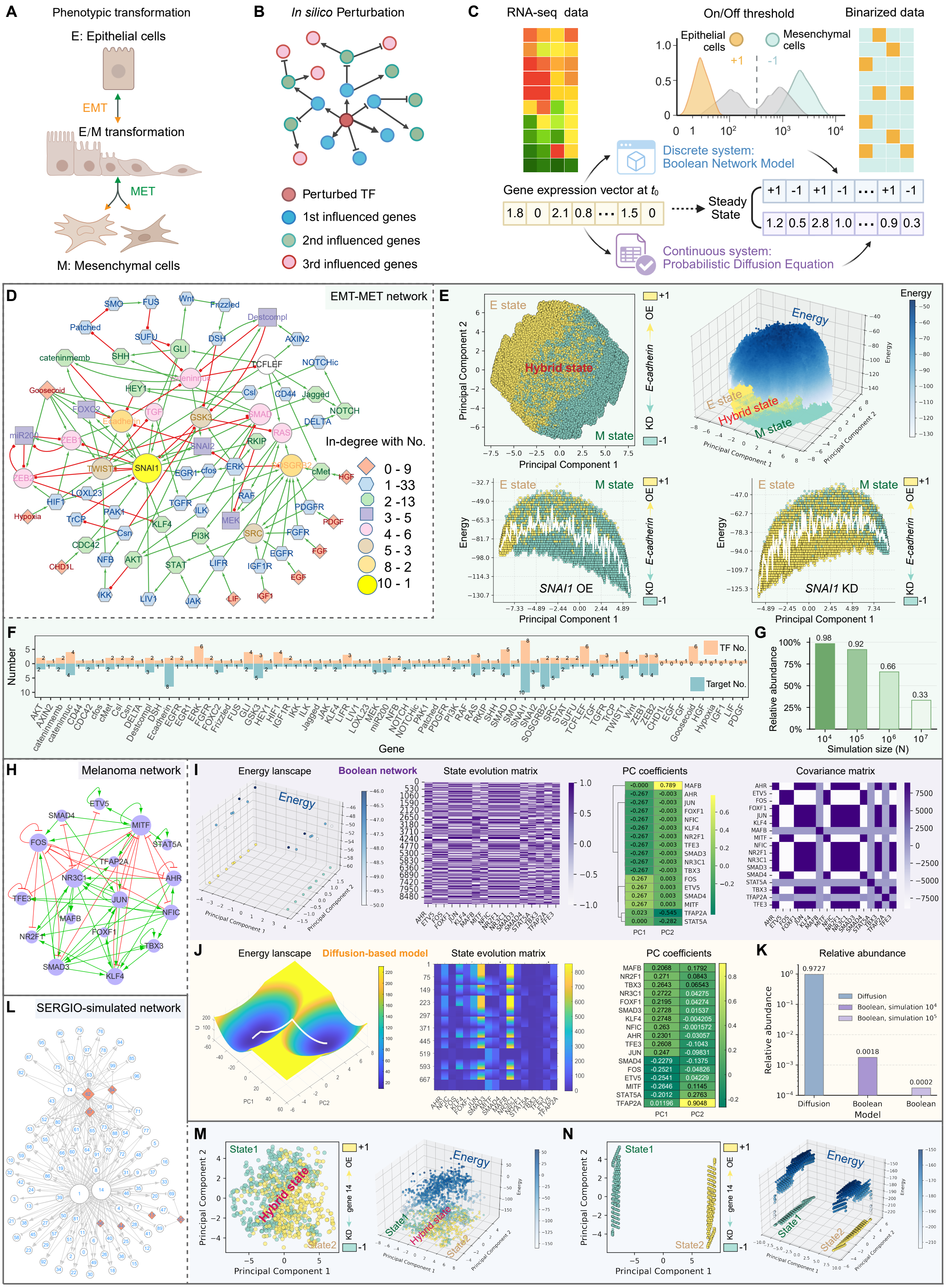

Fig. 1: *In silico* perturbation framework and energy landscape analysis of gene regulatory networks to study cell phenotypic transitions via stable state attractors. (Captions continue on the next page)

Fig. 1: **(A)** Schematic of phenotypic transitions between epithelial and mesenchymal cell states through epithelial-mesenchymal transition (EMT) and mesenchymal-epithelial transition (MET). **(B)** *In silico* perturbation of a gene regulatory network, showing the perturbed transcription factor (TF) and its first-, second-, and third-order downstream influenced genes. **(C)** Overview of the *in silico* perturbation workflow for discrete and continuous dynamical systems, implemented using a BN model and probabilistic diffusion equations, respectively. In the discrete setting, RNA-seq data are binarized using on/off thresholds defined from epithelial (E) and mesenchymal (M) reference states. In the continuous setting, the gene expression vector at t_0 is evolved under probabilistic diffusion equations to reach a steady state. **(D)** EMT-MET regulatory network in hepatocellular carcinoma (HCC). **(E)** State-space and energy-landscape analysis of the EMT-MET network. Top left, principal component analysis (PCA) of 10^6 steady states, showing the E, M and intermediate hybrid regions; colour denotes the *E-cadherin* state. Top right, 3D reconstruction of the corresponding EMT energy landscape, in which higher-energy states are concentrated in the transition region. Bottom, 1D PCA projections showing that *SNAI1* overexpression (OE) biases the landscape towards the M state, whereas *SNAI1* knockdown (KD) biases it towards the E state. **(F)** Number of TF regulators and downstream targets for each gene in the EMT-MET network. **(G)** Relative abundance of steady states in the BN model across increasing numbers of initial conditions (N). Relative abundance is defined as the number of unique steady states divided by the number of initial conditions that converge to any steady state. **(H)** Gene regulatory network in melanoma. **(I)** BN-model results for the melanoma network, including the energy landscape and the corresponding state-evolution matrix, principal component (PC) loading coefficients and covariance matrix. **(J)** Diffusion-model results for the melanoma network, including the energy landscape, and the corresponding state-evolution matrix that describes the node expression trajectory matrix generated during diffusion-based model simulation over time, and the heatmap of loading coefficients of each gene in the PC1 and PC2 directions after PCA. **(K)** Comparison of relative abundance between the diffusion model and the BN model; for the BN model, results from two simulation sizes are shown. **(L)** Denoised signed interaction network generated using SERGIO. **(M-N)** PCA projections of 10^4 steady states and the corresponding energy landscapes obtained from Langevin dynamics (**M**) and from the BN model initialized from random states (**N**).

high-dimensional state space into distinct attractor regions (**Fig.** 1E). Steady states were classified as E, M, or hybrid states according to the expression state of E-cadherin (*CDH1*). Mapping state abundance to the pseudo-energy defined by Eq. (7) revealed a rugged landscape, in which E and M states occupied deeper basins, whereas hybrid E/M states were enriched in elevated intermediate regions (**Fig.** 1E, top right; **Supp. Fig.** S1), consistent with their interpretation as metastable transition states. We next assessed how individual regulators reshape this landscape by fixing each node/gene i to $+1$ or -1, corresponding to OE or KD, respectively. OE of *SNAI1*, a canonical EMT inducer, shifted the 1D pseudo-energy profile towards the M basin, whereas *SNAI1* KD biased the system towards the E state (**Fig.** 1E, bottom). Thus, perturbation of a single TF can redirect attractor occupancy by reshaping the global landscape, indicating that the Boolean framework captures the essential regulatory constraints underlying EMT/MET plasticity [22]. The topological features of the EMT-MET network further support this view (**Fig.** 1F). The circuit comprises 72 nodes and 139 signed interactions, with an average degree of 3.694, a diameter of 13, and a radius of 7, indicating a sparse but globally connected architecture (**Fig.** 1D; **Tab.** 1). Such an organization allows local perturbations to propagate through multi-step regulatory cascades and generate system-wide changes in attractor structure. Consistent with this, the in-degree distribution shows that most nodes receive a moderate number of inputs (3-5), whereas only a few exhibit high in-degree (8-10; **Fig.** 1F). These highly connected nodes are therefore well positioned to integrate upstream signals and exert disproportionate control over cell-state transitions, making them plausible master regulators of E-M plasticity. Finally, we examined the convergence of the estimated relative abundance of attractor states during the time-evolution process (see Supp. Methods; **Fig.** 1G). As the number of simulations increased from 10^4 to 10^7, the estimates became progressively more stable, supporting the robustness of the reconstructed landscape. Finally, we examined the tendency of the relative abundance during the time-evolution simulations (see Supp. Methods), defined as the fraction of sampled random initial conditions associated with distinct steady states. As the number of simulations increased from 10^4 to 10^7 ($< 2^7 \sim 4.72 \times 10^{21}$), this abundance decreased but was higher than 25%, supporting the robustness of the reconstructed landscape (**Supp. Tab.** S1).

Landscape for melanoma gene regulatory network Phenotypic heterogeneity in melanoma is closely associated with drug resistance, relapse and therapeutic failure [34]. In addition to the canonical proliferative and invasive states, previous work identified a drug-tolerant, hyperpigmented state marked by high expression of the lineage regulator *MITF* and its pigmentation targets (*PMEL*, *TYR* and *MLANA*). To dissect the regulatory logic and state-transition trajectories underlying these phenotypes, Pillai *et al.* [34] analysed a 17-node GRN (**Fig.** 1H) using RACIPE [35] and reconstructed a diffusion-based energy landscape (Eq. (3)) that accounts for four melanoma phenotypes [23]. Here, we analysed the same network using both the BN model (Eq. (4)) and the diffusion-based model defined in Eq. (3). As in the EMT-MET analysis, we performed 10^4 BN simulations from random initial conditions and classified the resulting states using *MITF* activity, which distinguishes proliferative from hyperpigmented programmes [23]. In contrast to the EMT-MET network, the melanoma GRN has a distinct topology that strongly constrains Boolean dynamics. BN simulations therefore collapsed onto only 16 unique attractors despite extensive sampling of initial conditions (**Supp. Tab**. S2). This degeneracy is evident in the state-evolution and covariance matrices and produces a fragmented pseudo-energy landscape composed of sparse, isolated states rather than smooth basins (**Fig**. 1I). Among the $8,998$ simulations that converged to steady states, only 16 unique attractors were identified, and these could be grouped into five recurrent classes of node-activity patterns were observed (class 1: $\{6\}$; class 2: $\{13\}$; class 3: $\{15\}$; class 4: $\{1, 2, 7, 12\}$; class 5: $\{0, 3, 4, 5, 8, 9, 10, 11, 14, 16\}$), indicating that most genes remain locked in either the OE ($+1$) or KD (-1) state under the BN dynamics. Accordingly, the steady-state matrix has rank 5, suggesting that the emergent attractor structure is dominated by network topology. By contrast, the diffusion model evolves continuous gene-expression vectors under Eq. (3) and is not limited by this topological discretization. Under the selected parameterization (**Supp. Tab**. S3-S5), the system relaxed to a dense ensemble of steady states, enabling reconstruction of a smooth energy landscape (**Fig**. 1J). Consistent with this difference, the relative abundance of steady states, defined as the number of unique steady states divided by the number of initial conditions that converge to any steady state, was 0.9727 for the diffusion model, compared with 0.0018 and 0.0002 for BN simulations initiated from 10^4 and 10^5 states, respectively (**Fig**. 1K; **Supp. Fig**. S2). Thus, for the melanoma GRN, the diffusion model more effectively captures the continuous organisation of the state space and yields an interpretable energy landscape (**Supp. Fig**. S3), whereas the BN model collapses the dynamics into a small set of recurrent attractors imposed primarily by network topology.

Landscape of a signed interaction network in a SERGIO-simulated dataset To further compare discrete and continuous dynamical formalisms for the identification of steady-state attractors, we generated a synthetic single-cell expression dataset using the diffusion-based model (Eq. (3)) implemented in the SERGIO simulator [32], based on the ground-truth signed GRN (**Fig**. 1L). Specifically, we simulated a noise-free steady-state expression matrix simulated with `sergio(number_genes=100, number_bins=3, number_sc=300, noise_params=1, decays=0.8, sampling_state=15, noise_type='dpd')`, resulting in three cell types (300 cells per type) across 100 genes (7 TFs and 93 TGs) connected by 258 regulatory edges. To evaluate the BN model (Eq. (4)), we binarized the continuous expression profiles as shown in **Fig**. 1C. For each gene, the mean expression across the three cell types was used as the binarization threshold, resulting in a binary gene expression matrix. Gene 14, which has the highest degree in the network, was empirically selected as the classifier. In the diffusion-derived steady-state data, projection onto 2D PC space revealed two terminal states bridged by a substantial hybrid/intermediate population (**Fig**. 1M, left). Consistently, the corresponding pseudo-energy landscape, calculated using Eq. (7), placed these intermediate cells in a higher-energy region between the two terminal basins (**Fig**. 1M, right), indicative of a continuous transition structure in state space. By contrast, BN simulations performed on the same ground-truth GRN (**Fig**. 1 L) from random initial conditions and iterated to steady state yielded only two sharply separated terminal states, with minimal occupancy of the intermediate region (**Fig**. 1N, left). The corresponding 1D pseudo-energy profile, calculated using Eq. (7), also failed to exhibit a meaningful topography (**Fig**. 1N, right). Moreover, we also try to increase the number of simulations to 10^5 and 10^6, but obtain very similar unsatisfactory results. These findings highlight the differences between continuous and discrete dynamical systems in steady-state attractor detection. Although the BN model performs well for the EMT-MET network with extensive simulations, the identification of distinct attractors remains highly sensitive to network structure.

Characterizing cell-state dynamics on energy landscape with Boolean network In diffusion-based systems, sensitivity primarily stems from model parameters, boundary conditions, and domain geometry, all of which strongly affect the existence and properties of stationary solutions. In fact, for BN models, steady-state behavior depends not only on network topology, update rules (synchronous vs. asynchronous), and initial conditions, but also on the emergence of "frozen components"-nodes locked in fixed states [36]. Crucially, these frozen components smooth the underlying energy landscape, thereby driving the network towards orderly dynamics and facilitating adaptive evolution. As shown in **Fig**. 1 F, most nodes in the EMT-MET network (**Fig**. 1 D) have moderate in-degree values (from 3 to 5), while only a few exhibit high connectivity (from 8 to 10). These observations indicate that effective BN modeling requires a network of appropriate complexity; networks that are too simple or too complex may fail to capture diverse attractor states. Ideally, an optimized GRN should avoid redundancy, allowing for a rich set of diverse stable attractors and facilitating identification of minimal driver node sets that control long-term BN dynamics. Under these conditions, BN models can efficiently capture biologically meaningful cell states or transitions, avoiding the extensive parameter tuning and computational demands typical of diffusion-based approaches.

In this study, we utilize pseudo-energy to quantify cell-state dynamics under single-node perturbations. Specifically, we measure the energy associated with long-term network configurations following targeted perturbations, implemented by fixing the value of a node while initializing all others to maximal uncertainty. High pseudo-energy indicates the potential for state transitions, while low values suggest restricted accessibility to other states. Using this framework, we successfully reproduce the established E/M hybrid state in the EMT-MET network, as well as the *SNAI1* gene expression pattern associated with EMT plasticity (**Fig.** 1 E). Recent work by Parmer et al. [37] shows that the relative size of the optimal driver set in a network is determined solely by network degree, rather than overall network size or other structural features. Notably, nodes with zero in-degree must be included in the driver set, while those with high degree centrality are more likely to belong to minimal driver sets. Coincidentally, Huang et al. [38] introduced a network-structural method for identifying indicators that distinguish multiple steady states in chemical reaction networks, demonstrating that, using topological degree theory, certain subnetworks are sufficient to uniquely determine all species concentrations. Collectively, these advances highlight the growing importance of integrating structural and dynamical perspectives to better understand and control complex biological networks.

Conclusion

In summary, we compared steady-state detection and energy-landscape analysis in Boolean networks (BNs) and probabilistic diffusion equations (diffusion-based models). These two classes of models pose fundamentally different challenges, arising from their discrete versus continuous state spaces, the types of steady states they admit (point attractors and cycles versus stationary distributions and metastable states), and the computational techniques required for their analysis. BNs are limited by combinatorial state-space explosion and strong topology-dependent attractor search, whereas diffusion-based formulations require sophisticated analytical and numerical methods to resolve high-dimensional stationary solutions. By systematically examining how BN- and diffusion-based frameworks differ in attractor structure, sensitivity to network topology, and responses to perturbations on energy landscapes, our study clarifies when each modeling paradigm is most appropriate for studying gene regulatory dynamics. This comparison highlights practical trade-offs among computational cost, parameter sensitivity, and robustness to noise, and underscores the importance to align modeling choices with specific biological questions.

Recent methodological advances, including deep-learning–based landscape construction (for example, EPR-Net) and neural ODE or flow-matching approaches, provide powerful tools for learning non-equilibrium potential landscapes and continuous-time cell-state dynamics from high-dimensional data. These techniques can be applied to continuous stochastic models and, indirectly, to effective dynamics inferred from BN descriptions, thereby offering a bridge between mechanistic GRN-based modeling and data-driven landscape inference. Notably, our findings confirmed that the identification of influential attractors is highly dependent on the specific nature of network dynamics. As no unified theoretical framework yet exists to determine which types of networks, such as those defined by particular in-degree or out-degree constraints for certain nodes, are optimal for quantifying cell-state dynamics on energy landscapes via BN models, our work helps delineate this design space and motivates future efforts to systematically optimize GRN structures and integrate discrete and continuous descriptions in multiscale models of cell-state transitions.

Declarations

- **Author contributions**
  Lingyu Li: Conceptualization; investigation; methodology; data curation; validation; visualization; writing–original draft; writing–review and editing. **Liangjie Sun**: Methodology; writing–original draft; writing–review and editing. **Shumin Li**: Investigation; Writing–original draft; writing–review and editing. **Wai-Ki Ching**: Conceptualization; project administration; supervision; funding acquisition; writing–review and editing. **Zhi-Ping Liu**: Conceptualization; Project administration; supervision; funding acquisition; writing–review and editing.

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- **Conflict of interest interests**
  The authors declare no conflicts of interest.

- **Data availability statement**
  The EMT-MET interaction network in hepatocellular carcinoma (HCC) was downloaded from https://github.com/ComplexityBiosystems/bmodel/blob/master/bmodel/assets/topo_files/EMT_MET.topo, The gene regulatory network in the melanoma tumor was downloaded from https://github.com/csbBSSE/Melanoma/blob/main/melanoma.topo, The signed interaction network in SERGIO was simulated using the SERGIO package (https://github.com/PayamDiba/SERGIO).

- **Code availability statement**
  All Python and Matlab scripts, along with the input and output data used in this study, are available in the GitHub repository at https://github.com/zpliulab/CellLand. All data analyzed in this work are available through the figshare link: https://figshare.com/articles/dataset/CellLand_supplementary_data/32055507.

References

[1] Bunne, C., Roohani, Y., Rosen, Y., Gupta, A., Zhang, X., Roed, M., Alexandrov, T., AlQuraishi, M., Brennan, P., Burkhardt, D.B., *et al.*: How to build the virtual cell with artificial intelligence: Priorities and opportunities. Cell **187**(25), 7045–7063 (2024)

[2] Kamimoto, K., Stringa, B., Hoffmann, C.M., Jindal, K., Solnica-Krezel, L., Morris, S.A.: Dissecting cell identity via network inference and in silico gene perturbation. Nature **614**(7949), 742–751 (2023)

[3] Qiu, X., Zhang, Y., Martin-Rufino, J.D., Weng, C., Hosseinzadeh, S., Yang, D., Pogson, A.N., Hein, M.Y., Min, K.H.J., Wang, L., *et al.*: Mapping transcriptomic vector fields of single cells. Cell **185**(4), 690–711 (2022)

[4] Peng, Q., Zhou, P., Li, T.: stVCR: spatiotemporal dynamics of single cells. Nature Methods, 1–12 (2026)

[5] Wang, J., Zhang, K., Xu, L., Wang, E.: Quantifying the Waddington landscape and biological paths for development and differentiation. Proceedings of the National Academy of Sciences **108**(20), 8257–8262 (2011)

[6] Huang, S., Eichler, G., Bar-Yam, Y., Ingber, D.E.: Cell fates as high-dimensional attractor states of a complex gene regulatory network. Physical Review Letters **94**(12), 128701 (2005)

[7] Trinh, V.-G., Park, K.H., Pastva, S., Rozum, J.C.: Mapping the attractor landscape of Boolean networks with biobalm. Bioinformatics **41**(5), 280 (2025)

[8] Kauffman, S.A.: Metabolic stability and epigenesis in randomly constructed genetic nets. Journal of Theoretical Biology **22**(3), 437–467 (1969)

[9] Ching, W.-K., Zhang, S., Ng, M.K., Akutsu, T.: An approximation method for solving the steady-state probability distribution of probabilistic Boolean networks. Bioinformatics **23**(12), 1511–1518 (2007)

[10] Trairatphisan, P., Mizera, A., Pang, J., Tantar, A.A., Schneider, J., Sauter, T.: Recent development and biomedical applications of probabilistic boolean networks. Cell Communication and Signaling **11**(1), 46 (2013)

[11] Kauffman, S.A.: The origins of order: Self-organization and selection in evolution (1993)

[12] Bornholdt, S.: Boolean network models of cellular regulation: prospects and limitations. Journal of the Royal Society interface **5**(suppl_1), 85–94 (2008)

[13] Li, L., Sun, L., Chen, G., Wong, C.-W., Ching, W.-K., Liu, Z.-P.: LogBTF: gene regulatory network inference using Boolean threshold network model from single-cell gene expression data. Bioinformatics **39**(5), 256 (2023)

[14] Li, Z., Wang, T., Liu, P., Huang, Y.: SpatialDM for rapid identification of spatially co-expressed ligand-receptor and revealing cell–cell communication patterns. Nature Communications **14**(1), 3995 (2023)

[15] Jin, K., Chen, Z., Yu, X., Yuan, M., Schroeder, A., Dumoulin, B., Liu, Y., Wang, L., Park, J.H., Hwang, T.H., et al.: Multiscale confidence quantification for virtual spatial transcriptomics with UTOPIA. BioRxiv, 2026–03 (2026) https://doi.org/10.64898/2026.03.01.708850

[16] Li, L., Wang, T., Liang, Z., Yu, H., Ma, S., Yu, L., Huang, Y.: FineST: Contrastive learning integrates histology and spatial transcriptomics for nuclei-resolved ligand-receptor analysis. Nature Communications (2026) https://doi.org/10.1038/s41467-026-70528-7

[17] Wang, T., Li, L., Huang, Y.: SparseAEH: Scalable autoregressive expression histology discovery in spatial transcriptomics via sparse Gaussian kernels. bioRxiv, 2026–01 (2026) https://doi.org/10.64898/2026.01.22.701193

[18] Shi, J., Aihara, K., Li, T., Chen, L.: Energy landscape decomposition for cell differentiation with proliferation effect. National Science Review **9**(8), 116 (2022)

[19] Zhou, P., Li, T.: Construction of the landscape for multi-stable systems: Potential landscape, quasi-potential, A-type integral and beyond. The Journal of Chemical Physics **144**(9) (2016)

[20] Jung, I., Hopper, C., Jang, S.-H., Yeo, H., Cho, K.-H.: Reverse control of biological networks to restore phenotype landscapes. Science Advances **11**(34), 3995 (2025)

[21] La Porta, C.A., Zapperi, S.: Phase transitions in cell migration. Nature Reviews Physics **2**(10), 516–517 (2020)

[22] Font-Clos, F., Zapperi, S., La Porta, C.A.: Topography of epithelial–mesenchymal plasticity. Proceedings of the National Academy of Sciences **115**(23), 5902–5907 (2018)

[23] Pillai, M., Chen, Z., Jolly, M.K., Li, C.: Quantitative landscapes reveal trajectories of cell-state transitions associated with drug resistance in melanoma. iScience **25**(12), 105499 (2022)

[24] Kang, X., Li, C.: A dimension reduction approach for energy landscape: identifying intermediate states in metabolism-EMT network. Advanced Science **8**(10), 2003133 (2021)

[25] Zhao, Y., Zhang, W., Li, T.: EPR-Net: constructing a non-equilibrium potential landscape via a variational force projection formulation. National Science Review **11**(7), 052 (2024)

[26] Richter, T., Wang, W., Palma, A., Theis, F.J.: Generative models of cell dynamics: from Neural ODEs to flow matching. Communications Biology **9**(352), 1–10 (2026)

[27] Zhu, J., Zhang, Z., Sun, Y., Dai, H., Wen, H., Zhou, P., Chen, L.: Generating and navigating single cell dynamics via a geodesic bridge between nonlinear transcriptional and linear latent manifolds. bioRxiv, 2026–03 (2026) https://doi.org/10.64898/2026.03.31.715478

[28] Kramers, H.A.: Brownian motion in a field of force and the diffusion model of chemical reactions.

Physica **7**(4), 284–304 (1940)

[29] D'Souza, R.M., Di Bernardo, M., Liu, Y.-Y.: Controlling complex networks with complex nodes. Nature Reviews Physics **5**(4), 250–262 (2023)

[30] Risken, H.: Fokker-planck equation, 63–95 (1989)

[31] Yin, J., Wang, Y., Chen, J.Z., Zhang, P., Zhang, L.: Construction of a pathway map on a complicated energy landscape. Physical Review Letters **124**(9), 090601 (2020)

[32] Dibaeinia, P., Sinha, S.: SERGIO: A single-cell expression simulator guided by gene regulatory networks. Cell Systems **11**(3), 252–271 (2020)

[33] Peng, R., Gao, Y., Chen, Y., Wang, J.: CellProphet: Dissecting virtual cell differentiation through AI-powered dynamic gene regulatory network inference. bioRxiv (2025) https://doi.org/10.1101/2025.02.05.636766

[34] Pillai, M., Jolly, M.K.: Systems-level network modeling deciphers the master regulators of phenotypic plasticity and heterogeneity in melanoma. iScience **24**(10), 103111 (2021)

[35] Huang, B., Lu, M., Jia, D., Ben-Jacob, E., Levine, H., Onuchic, J.N.: Interrogating the topological robustness of gene regulatory circuits by randomization. PLoS Computational Biology **13**(3), 1005456 (2017)

[36] Kauffman, S.A.: Requirements for evolvability in complex systems: orderly dynamics and frozen components. Physica D: Nonlinear Phenomena **42**(1-3), 135–152 (1990)

[37] Parmer, T., Rocha, L.M., Radicchi, F.: Influence maximization in Boolean networks. Nature Communications **13**(1), 3457 (2022)

[38] Huang, Y.-J., Mochizuki, A., Okada, T.: Identifying phenotype-indicative molecules from the structure of biochemical reaction networks. bioRxiv, 2025–10 (2025) https://doi.org/10.1101/2025.10.21.683796